

SOLUTIONS

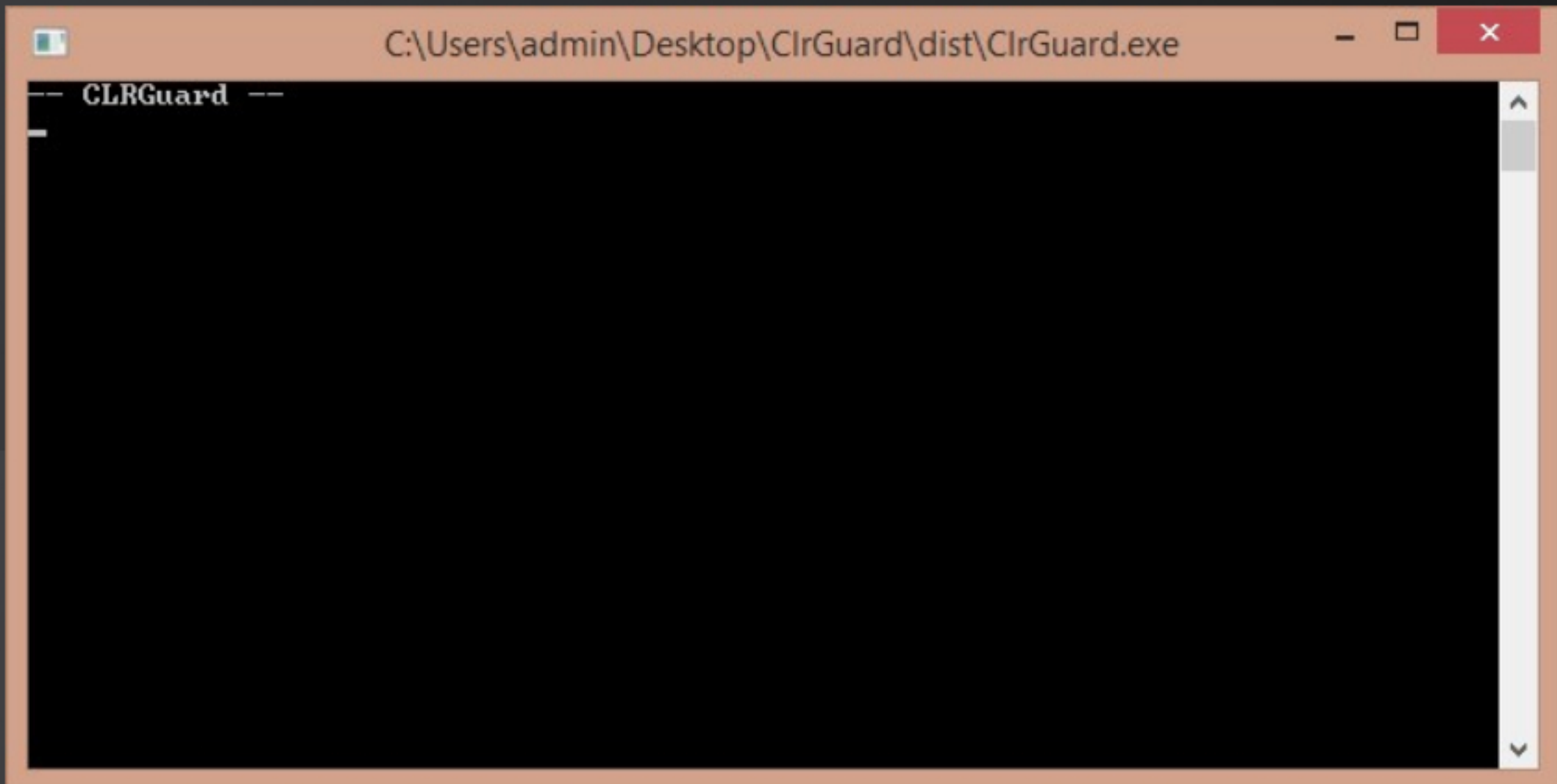
Below, you can find solutions for each task. Remember though, that you can follow your own strategy, which may be different from the one explained in the following lab.

Task 1. Get familiar with ClrGuard

Execute *ClrGuard.exe* as administrator.

ClrGuard.exe is located at: *C:\Users\admin\Desktop\ClrGuard\dist*

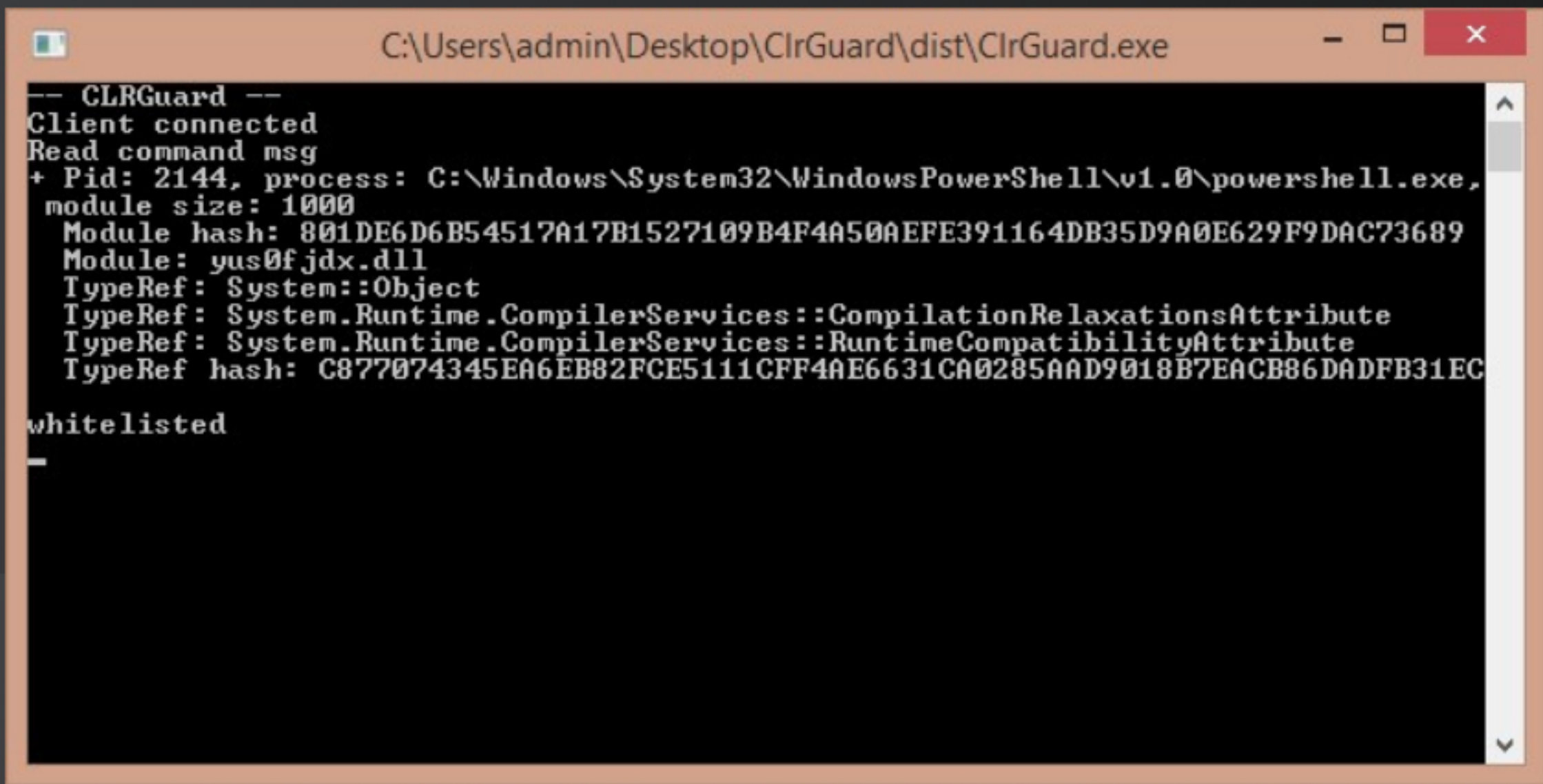
You should see the below.



Now, double-click *fp.bat*.

fp.bat is located at: *C:\Users\admin\Desktop\ClrGuard\Testing\scripts*

You should see the below.

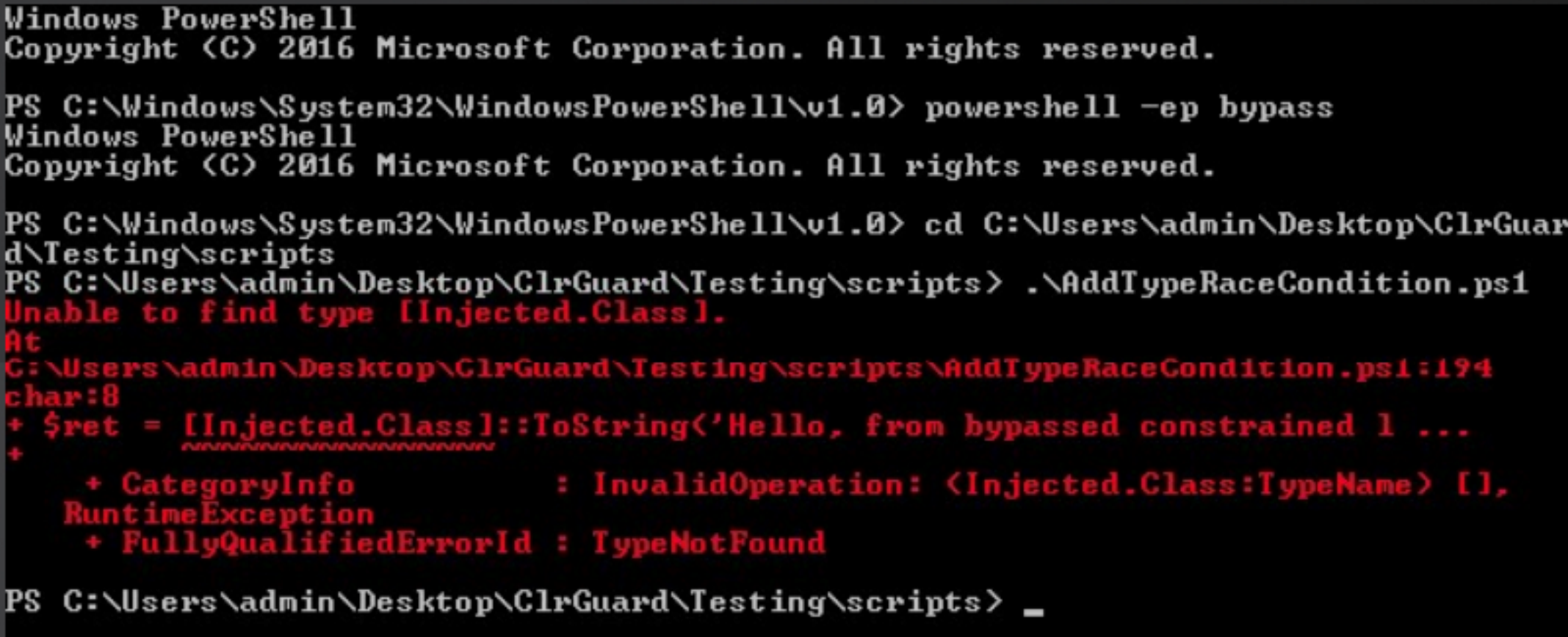


The module being loaded is whitelisted by ClrGuard. Assembly loading can also be performed for legitimate purposes!

Now, execute the below.

Click the PowerShell icon that is pinned on the taskbar and execute the below.

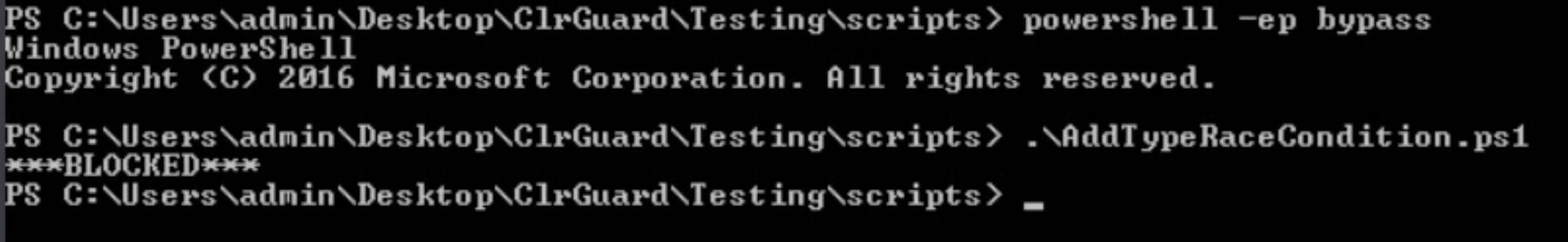
```
powershell -ep bypass
cd C:\Users\admin\Desktop\ClrGuard\Testing\scripts
.\AddTypeRaceCondition.ps1
```



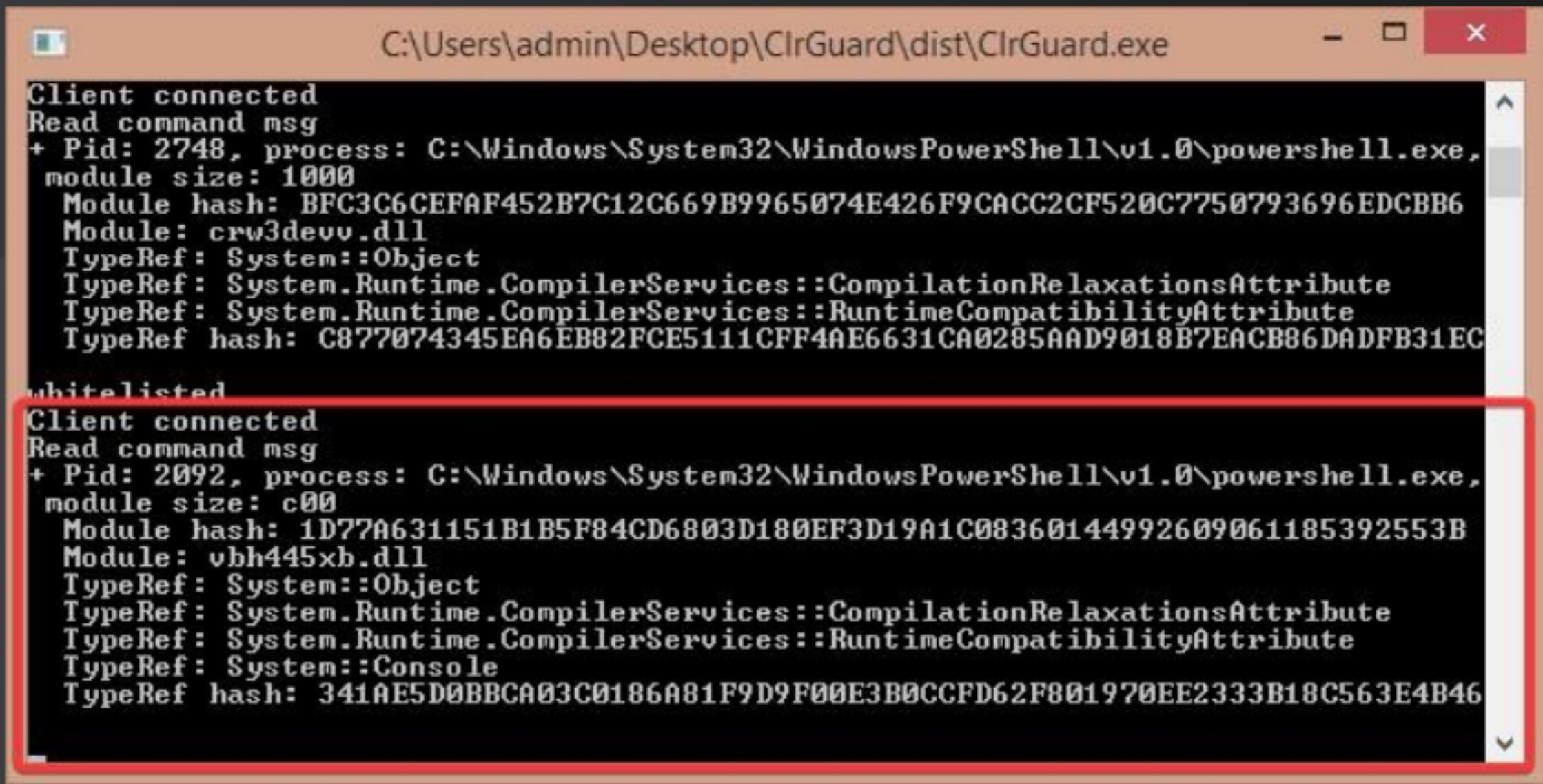
Note: If you receive the above error, Please re-run all the commands again.

fp.bat is located at: *C:\Users\admin\Desktop\ClrGuard\Testing\scripts*

You should see the below.



Inside ClrGuard, you should see something similar to the below.



The specific .net assembly loading was blocked by ClrGuard. Notice that there is no "whitelisted" string in the second entry.

Task 2. Emulate execute-assembly activity and use ClrGuard to proactively detect it

Let's now better emulate the execute-assembly command/functionality of Cobalt Strike.

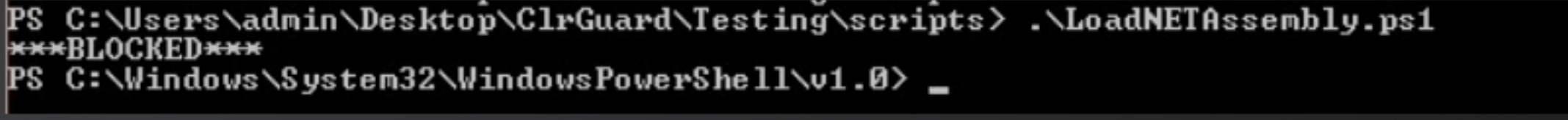
In the *scripts* directory, there is a PowerShell script called *LoadNETAssembly.ps1*.

To execute it, perform the following.

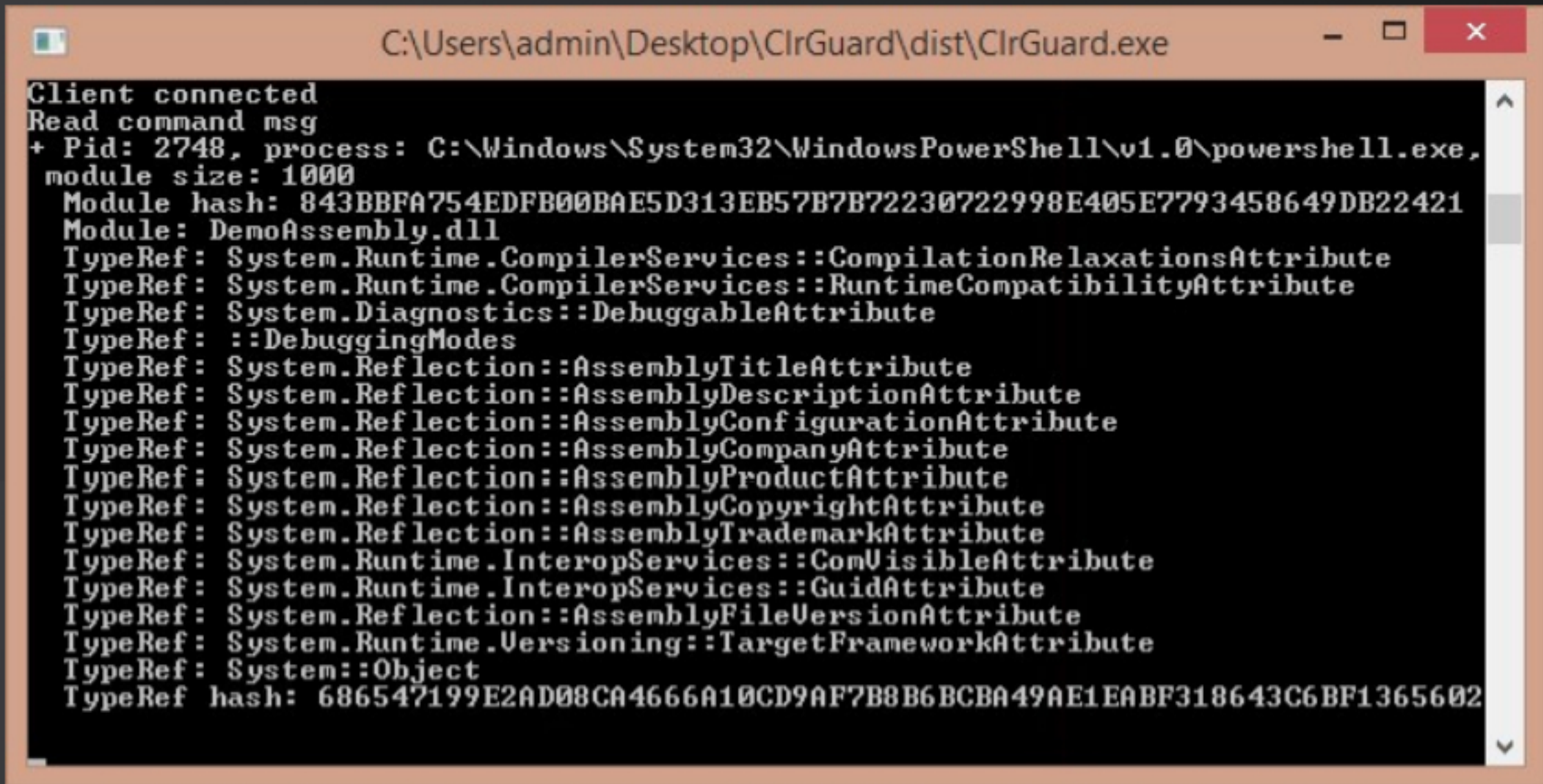
Click the PowerShell icon that is pinned on the taskbar and execute:

```
powershell -ep bypass
cd C:\Users\admin\Desktop\ClrGuard\Testing\scripts
.\LoadNETAssembly.ps1
```

You should see the below.



Inside ClrGuard you should see something similar to the below.



Notice that there is no "whitelisted" string in the last entry.

Note that there are *execute-assembly* variations included in multiple attacking frameworks. The detection method covered doesn't work only against Cobalt Strike.

Not Running

For the best experience, choose the region closest to you. Then, start the lab to begin.

MARK UNFINISHED

Select the region closest to you

US-East

Keyboard layout: USEnglish (US)

Start lab